Also included in the chat were the csv files from the Schedules and Teams folders, as well as screenshots of my UML diagram and ERD, and the create table statements. Opus 4.6 Extended Thinking was used.

Prompt 1:

I need sample data for each of the tables defined in the attached create table sql statements. They very closely model FRC. Attached are the UML diagram, ERD, and multiple csv files. The csv files for the competition named "Greater Pittsburgh Regional" are prefixed with "Pitt", "Buckeye Regional" is prefixed with "Buckeye", and "Finger Lakes Regional" is prefixed with "Finger". Each competition name has a csv file for teams and schedules for the games from years 2023, 2024, and 2025.

The sample data for the "Competition" table therefore should include 9 competitions: each name for each year from 2023, 2024, 2025. Research online to find the rest of the attributes.

There are only 3 games to be included in the sample data for "Game", the games from the three listed years. Research online to find the attributes.

The 18 csv files will come in handy for the "MatchPerformance" and "Team" tables:

To create data for the "Team" table: find all unique values for the field "team_number" across all csv files with the keyword "Teams". For each team number from this result, extract the team name, city, state, and country for the test data. Research online to find the team's rookie year, and make up a value for pointsLevel (make a random amount NULL).

For each year/game that the team attended a competition in, they will have a different robot according to the ERD. Use this information and other visible constraints to generate data for the "Robot" table. Make up values for robotName and aprilTags. For drivebase, choose one from this set - Swerve, Mecanum, 4-Wheel, 6-Wheel.

For the constrained system of 4 tables to describe each robot's scoring and pickup abilities as a subset of the possible values for the game it was built for:

2023 Possible Scoring Set - Cube1, Cube2, Cube3, Cone1, Cone2, Cone3

2023 Possible Pickup Set - CubeFloor, ConeFloor, CubeStation, ConeStation

2024 Possible Scoring Set - Amp, Speaker, Trap, Climb, Harmonize, Human

2024 Possible Pickup Set - Floor, Source

2025 Possible Scoring Set - AlgaeBarge, AlgaeProcessor, Coral1, Coral2, Coral3, Coral 4, ShallowHang, DeepHang, Human

2025 Possible Pickup Set - AlgaeFloor, AlgaeReef, CoralFloor, CoralStation

Generate random subsets of each for each robot.

Use the csv files again for MatchPerformance:

For each csv file with the keyword "Schedule", for each match and each robot extract the match number and alliance of the robot. Generate random values for the rest of the attributes. However, make sure that when you add all points across all categories (auto, endgame, teleop) for each robot in the alliance, that this result equals the alliance score present in the csv.

For "SpecialEvent":

For a random amount of match performances, add a random amount of events from this set - Died, Broke, Tipped, Foul (don't duplicate each individual type of event.

For "TeamMember":

gradClass must range from their team's rookie year to 2029. Each team must have at least one member. If their team's value for pointsLevel is NULL, their value for points is NULL. Otherwise, generate a random value. Also generate random values for the name, subteam, compjob, and ismentor.

WARNING:

Follow the totality of all constraints found in the ERD, UML, this prompt, and the create table statements.

EXPECTED OUTPUT:

For each table for the create table statements, generate a csv with the sample data, making sure to follow the exact schema. Before fully generating them however, display samples of each so I can validate some data.

Output 1:

Here are sample previews for all 13 tables. I'm showing 3–5 rows each so you can validate before I generate the full CSVs.

Competition (9 total rows)

competitionID	competitionName	week	year	city	state	country
1	Greater Pittsburgh Regional	6	2023	Pittsburgh	Pennsylvania	USA
2	Greater Pittsburgh Regional	1	2024	Pittsburgh	Pennsylvania	USA
3	Greater Pittsburgh Regional	4	2025	Pittsburgh	Pennsylvania	USA
4	Buckeye Regional	5	2023	Cleveland	Ohio	USA
5	Buckeye Regional	4	2024	Cleveland	Ohio	USA
…	…	…	…	…	…	…

Game (3 total rows)

gameName	theme	cargoCount
Charged Up	Energy	98
Crescendo	Arts	113
Reefscape	Ocean	138

Cargo counts: 2023 = 54 cones + 44 cubes = 98. 2024 = 113 notes on field. 2025 = ~120 coral + 18 algae = 138. Let me know if you'd prefer different values.

CompetitionGame (9 total rows — one per competition)

competitionID	gameName
1	Charged Up
2	Crescendo
3	Reefscape
4	Charged Up
5	Crescendo
…	…

Team (180 total rows — sample of 5)

teamNumber	teamName	rookieYear	pointsLevel	city	state	country
48	Team E.L.I.T.E.	1998	3	Warren	Ohio	USA
120	Youth Tech Academy Red Dragons	1999	NULL	Cleveland	Ohio	USA

teamNumber	teamName	rookieYear	pointsLevel	city	state	country
1507	Warlocks	2005	7	Lockport	New York	USA
5740	Trojanators	2016	NULL	Bellevue	Ohio	USA
9767	Robo Lions	2025	2	Cleveland	Ohio	USA

Team name, city, state, country come directly from the CSVs. Rookie year is researched (low numbers like 48 → 1998 confirmed via TBA; higher numbers estimated from number ranges). About 30% of pointsLevel values will be NULL.

Robot (one per team per game-year they attended — sample of 5)

robotID	gameName	teamNumber	robotName	aprilTags	drivebase
1	Charged Up	48	Thunderclap	Y	Swerve
2	Charged Up	59	Rampage	N	6-Wheel
3	Charged Up	117	Iron Fury	Y	Mecanum
4	Crescendo	48	Thunderclap II	Y	Swerve
5	Reefscape	48	Thunderclap III	Y	Swerve

A team that attended competitions in all 3 years gets 3 robots. A team that only appeared in 2024 files gets 1 robot. Robot names are fabricated. Drivebase is randomly selected from {Swerve, Mecanum, 4-Wheel, 6-Wheel}.

CompetitionRobot (sample of 5)

competitionID	robotID
1	2
1	25
1	47
4	1
7	1

This links each robot to every competition it played at. E.g., team 48's 2023 robot (robotID 1) appears at both Pitt 2023 (compID 1) and Buckeye 2023 (compID 4) if they attended both.

GameScoring (19 total rows)

gameName	ability
Charged Up	Cube1
Charged Up	Cube2
Charged Up	Cube3
Charged Up	Cone1
Charged Up	Cone2
Charged Up	Cone3
Crescendo	Amp
Crescendo	Speaker
…	…
Reefscape	Coral4
Reefscape	DeepHang
Reefscape	Human

GamePickup (10 total rows)

gameName	ability
Charged Up	CubeFloor
Charged Up	ConeFloor
Charged Up	CubeStation
Charged Up	ConeStation
Crescendo	Floor
Crescendo	Source
Reefscape	AlgaeFloor
Reefscape	AlgaeReef
Reefscape	CoralFloor
Reefscape	CoralStation

RobotScoring (random subsets — sample of 5)

robotID	gameName	ability
1	Charged Up	Cube1
1	Charged Up	Cube2
1	Charged Up	Cone1
4	Crescendo	Amp
4	Crescendo	Speaker

Each robot gets a random subset (at least 1) of the scoring abilities for its game.

RobotPickup (random subsets — sample of 5)

robotID	gameName	ability
1	Charged Up	CubeFloor
1	Charged Up	ConeStation
4	Crescendo	Floor
4	Crescendo	Source
5	Reefscape	CoralStation

Same logic — random subset (at least 1) of pickup abilities for that robot's game.

MatchPerformance (qual matches only, 6 rows per match — sample from Pitt 2023 match 1)

matchID	competitionID	robotID	match	alliance	auto	teleop	endgame	rankingPoints	penaltyCard
1	1	12	1	Red	12	25	5	2	NULL
2	1	45	1	Red	8	20	3	2	NULL
3	1	102	1	Red	5	12	3	2	NULL
4	1	30	1	Blue	15	30	10	3	NULL
5	1	110	1	Blue	10	25	5	3	NULL
6	1	22	1	Blue	8	12	5	3	Yellow

This corresponds to CSV row: red1=59, red2=2872, red3=8140, blue1=1708, blue2=8705, blue3=1591, red_score=93, blue_score=120. The Red robots' totals: (12+25+5) + (8+20+3) + (5+12+3) = 42+31+20 = **93** ✓. The Blue robots' totals: (15+30+10) + (10+25+5) + (8+12+5) = 55+40+25 = **120** ✓. Ranking points are the same for all 3 robots on an alliance in a given match. Most penaltyCard values are NULL; a small percentage are Yellow or Red.

SpecialEvent (sample of 4)

eventID	matchID	event
1	3	Died
2	3	Tipped
3	17	Foul
4	52	Broke

About 10–15% of match performances get 1–3 events, drawn without replacement from {Died, Broke, Tipped, Foul}.

TeamMember (avg 20 per team, max 40 — sample of 5)

memberID	teamNumber	memberName	gradClass	points	subTeam	compJob	isMentor
1	48	Marcus Chen	2025	14	Build	Driver	N
2	48	Sophia Rodriguez	2026	8	Programming	Scouter	N
3	48	James Patterson	NULL	NULL	NULL	NULL	Y
4	48	Aiden Brooks	2027	5	CAD	Pit Crew	N
5	48	Dr. Linda Foster	NULL	NULL	NULL	Y	

Key rules reflected here: Team 48 has pointsLevel = 3 (not NULL), so student members get random point values. Team 120 has pointsLevel = NULL, so all its members' points will be NULL. Mentors (isMentor = 'Y') have gradClass = NULL per your updated constraint. Student gradClass ranges from the team's rookie year to 2029. subTeam drawn from {Build, Programming, CAD, Business, Electrical, Strategy}. compJob drawn from {Driver, Operator, Human Player, Scouter, Pit Crew, Technician, Coach, Strategist}. Both subTeam and compJob can randomly be NULL.

Does everything look correct? Any values, ranges, or naming conventions you'd like me to adjust before I generate the full CSVs?

Prompt 2:

Game: For cargoCount, I meant types of cargo present in the game. ChargedUp would be 2, Crescendo would be 1, and Reefscape would be 2.

TeamMember: For mentors, make gradClass, points, and compJob null. Also, make note that there is a missing value in the last row in the sample, causing isMentor for memberID 5 to have no value at all, not even NULL.

After considering this, create each of the full csv files.

Output 2:

Got it — fixing both. Let me generate everything now.

Prompt 3;

Try generating them again, by yourself. Just like the samples, don't use a python script, just give me csv files only.

Output 3:

Got it — I'll generate everything behind the scenes and just give you the CSV files. This will take several steps due to the volume of data.Now let me run a few validation checks before handing them over.All validations pass — zero constraint violations across the board. Let me present the files.Here are all 13 CSV files. A summary of what was validated with zero violations:

- **Score sums**: All 741 qualification matches across 9 competitions — each alliance's auto+teleop+endgame totals across 3 robots match the red_score/blue_score from the original schedule CSVs
- **Mentor rules**: gradClass, points, and compJob are all NULL for every mentor
- **Points/pointsLevel linkage**: Students on teams with NULL pointsLevel have NULL points; students on teams with a pointsLevel have a generated value
- **Every team** has at least 1 member
- **No duplicate event types** per SpecialEvent matchID
- **Every (competition, robot) pair** in MatchPerformance exists in CompetitionRobot
- **All RobotScoring/RobotPickup abilities** reference valid GameScoring/GamePickup entries for the correct game